

DevOps Case Study

K8s Cluster (not expecting to have running cluster):

- We are asking you to create Kubernetes cluster creation config for kops which has multiple ig, mixed instance group and lifecycle (spot and ondemand).
- Have cluster autoscaler for all instance groups

Infra:

- Create deployment which contains Nginx and below web application running (within same pod) and sharing public files (css, js etc) through shared storage (not nfs).
- Expose application with creating service object.
- Implement auto scaling for deployment.
- Implementing/using basic configuration management with Ansible (to have application configs in Ansible)
- Use helm to render Kubernetes objects for re-using while creating new environments

Development:

- Develop a basic web application (using Python, Node.js or Go) to parse and process CSV files like attached format. When processing file you can just print content of the lines to browser.
- Web application should have basic interface to upload CSV and show previously processed files.
- Once CSV file processed upload it to the s3 storage.
- Waiting you to implement s3 glacier transition on s3 config.

Notes:

- You can use Minikube to deploy application locally
- You can use dockerhub to store Docker images
- You can use github to store application and infra codes

Attachment: soh.csv

Solutioning:

- Documentation and architecture diagram